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RESEARCH PAPER

Suppressive activities of lupeol on sepsis mouse model

Sanghee Cho[1] · Yun Jin Park[1] · Jinhee Lee[1] · Jong-Sup Bae[1]

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Abstract

Sepsis is a life-threatening condition triggered by the body's extreme response to an infection, leading to widespread inflammation, organ dysfunction, and potentially fatal complications. While lupeol, a significant phytosterol found in various herbal plants, has been considered as a potential anti-cancer agent, its anti-septic activities and underlying molecular mechanisms remain unclear. The aim of this study is to investigate the effects of lupeol on a cecal ligation and puncture (CLP)-induced septic mouse model. Animals were categorized into six groups: control, CLP-operated, CLP plus maslinic acid, and CLP plus lupeol (0.5, 1, or 2 mg/kg). The assessment included survival rate, body weight changes, inflammatory cytokines, and histological analyses. Additionally, human endothelial cells were stimulated with high mobility group box1 (HMGB1) protein and lupeol, with cell viability determined. Inflammatory markers and gene expression were evaluated through enzymelinked immunosorbent assay and Western blot analysis, respectively. After CLP surgery, the group treated with lupeol showed improved survival rates and body weight compared to the untreated control group. Lupeol treatment also decreased levels of tumor necrosis factor (TNF)-α, interleukin-1β, nitric oxide, and cytokines associated with kidney inflammation. When administered to HMGB1-activated cells, lupeol reduced the expression of Toll-like receptor 4 and TNF-α, while simultaneously activating phosphoinositide 3-kinase/AKT signaling to enhance cell survival. In conclusion, lupeol demonstrated anti-inflammatory properties and conferred protective effects against CLP-induced sepsis, reinforcing cell survival in the face of septic responses.

Keywords Lupeol · Cecal ligation and puncture · Sepsis · Kidney injury · Inflammation

1 Introduction

Bombax ceiba (*Bombax malabarica*, Bombacaceae), also known as white silk cotton or semul, is a medicinal plant extensively cultivated in East Asia, encompassing regions such as Pakistan, India, China, and Vietnam [1, 2]. Renowned for its therapeutic properties, this plant is traditionally used to treat conditions such as diarrhea, fever, chronic inflammation, hepatitis, and contused wounds [1, 2]. Given its medicinal significance, numerous studies have been conducted on *Bombax ceiba* in recent years. Pharmacologically, the plant has demonstrated hepatoprotective qualities [1], hypotensive activity [2], and anti-inflammatory effects [3]. Lupeol (Fig. 1), a prominent phytosterol found in *Bombax ceiba*, is a pentacyclic triterpenoid known for its substantial biological activity against various human diseases [4, 5]. In cancer cells, lupeol exhibits anticancer effects by modulating the expression of interleukin (IL)-1β, 2, -4, -5, proteases, α-glucosidase, and nuclear factor (NF)-κB. It also induces cell death by altering the expression levels of B-cell leukemia (BCL)-2, Bcl-2-associated X protein (BAX), caspases, and the phosphoinositide 3-kinase (PI3K)-AKT-mTOR signaling pathway [6].

Sepsis, a clinical syndrome associated with systemic inflammation due to infection [7], is often marked by a "cytokine storm", a rapid release of proinflammatory cytokines, leading to multiorgan dysfunction and failure [8–10]. In 2020, there were an estimated 48.9 million sepsis cases worldwide, resulting in approximately 11 million sepsis-related deaths [11]. It is a systemic immune disorder caused by bacterial, fungal, or viral infections and is linked

Sanghee Cho and Yun Jin Park have contributed equally to this work.

✉ Jong-Sup Bae
baejs@knu.ac.kr

[1] College of Pharmacy, Research Institute of Pharmaceutical Sciences, Kyungpook National University, Daegu 41566, Korea

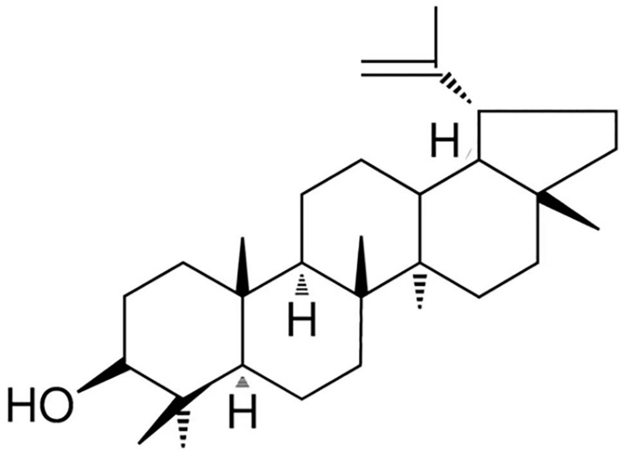

Fig. 1 The chemical structure of lupeol

to significant mortality rates (30–50% in-hospital mortality rates) [12]. Characterized by systemic inflammation termed "systemic inflammatory response syndrome" or SIRS, sepsis is increasingly responsible for prevalent illnesses and fatalities, particularly among older adults, immunocompromised individuals, and critically ill patients. As a result, sepsis has emerged as a significant global health and economic burden. Pattern recognition receptors play a role in innate immunity by identifying various pathogen-associated molecular patterns (PAMPs) and damage-associated molecular patterns (DAMPs) [13, 14]. The excessive production of PAMP and DAMP sensors during sepsis is implicated in multiple organ failure (MOF) [15–17]. Xigris, previously used as a treatment for sepsis, was withdrawn from the market in 2011 due to side effects and efficacy issues [18]. Since then, there have been no approved treatments for sepsis. Consequently, the development of alternative drugs for sepsis has become essential.

Cecal ligation and puncture (CLP) surgery stands as the predominant method for experimentally inducing sepsis in rodents and is currently acknowledged as the standard approach for sepsis research [19, 20]. Despite being developed over 40 years ago, the CLP model remains a realistic and widely utilized method for inducing polymicrobial sepsis, serving as a valuable tool for investigating sepsis mechanisms and potential treatments [19, 20]. Given the absence of studies exploring the association between lupeol (a phytosterol found in various herbal plants), and sepsis, our investigation aimed to elucidate the impact of lupeol on CLP-induced sepsis. Additionally, we suggest that the lupeol may offer anti-inflammatory effects against CLP-induced septic mice.

2 Materials and methods

2.1 Cell culture and reagents

Primary human umbilical vein endothelial cells (HUVECs) were procured from Cambrex Bio Science and cultivated following established protocols [21, 22]. Lupeol, 2,5-diphenyl-2H-tetrazolium bromide (MTT), dimethyl sulfoxide (DMSO), and maslinic acid (MA) were sourced from Sigma Chemical Co., with MA serving as a positive control [23–25]. Genetically modified human high mobility group box1 (HMGB1) was acquired from Abnova.

2.2 Animals and the CLP procedure

We purchased male C57BL/6 mice from Orient Bio Co., weighing an average of 27 g at 8–10 weeks of age. The mice were housed under standard animal care conditions with ad libitum access to food and water. All procedures were conducted following the guidelines outlined in the "Guide for the Care and Use of Laboratory Animals". Either moderate or severe CLP surgery was performed according to established protocols [26–28]. To induce sepsis, a midline incision of 2 cm was made, and the cecum was exposed adjacent to the ileum. Subsequently, the cecum was securely ligated at 5.0 mm from the cecal tip using 3.0 silk sutures, followed by a single puncture with a 22-gauge needle for moderate severity or two punctures for severe severity. A small amount of feces was gently extruded from the puncture site by compressing the cecum, and then returned to the peritoneal cavity. The abdominal incision was closed using 4.0 silk sutures. In the sham-operated group, the cecum was exposed but neither ligated nor punctured, and then returned to the peritoneal cavity. The cecum in the sham group was exposed without ligation or puncture. This experimental protocol was approved by the Institutional Animal Care and Use Committee of Kyungpook National University before the start of the study (IRB No. KNU 2022-113).

2.3 Experimental design

The mice were categorized into six groups, each comprising 20 individuals, as outlined: the control sham group, CLP group, CLP plus MA (2 mg/kg) group, and CLP plus lupeol groups with concentrations of 0.5, 1, or 2 mg/kg. Substances or saline solutions were administered intravenously 24 h post-CLP surgery. Blood samples were obtained from the orbital venous plexus 24 h after drug or saline administration, followed by centrifugation at 3000×*g* at 4 °C for 10 min. The resulting serum was then stored at – 80 °C.

2.4 Hematoxylin and eosin (H&E) staining and histopathological examination

After a 24 h period following drug administration, kidney specimens were preserved in 10% (v/v) phosphatebuffered formalin for 24 h and subsequently embedded in paraffin. Slices of 4 µm thickness were created using a microtome, subjected to H&E staining, and examined under a light microscope, following established procedures [27, 28].

2.5 Biochemical measurement

Levels of tumor necrosis factor (TNF)-α and IL-1β in kidney or liver tissues were measured using enzyme-linked immunosorbent assay (ELISA) kits from R&D Systems, following established protocols [29]. The concentration of nitric oxide (NO) in the kidneys was evaluated using the NO reductase method [30–32]. Measurement of phosphorylated PI3K levels was performed using ELISA kits obtained from MyBiosource.

2.6 Cell viability assay

After activating HUVECs with HMGB1 for 6 h, cells were treated with lupeol for 48 h, followed by estimation of cell viability using the MTT assay [29, 32].

2.7 Western blot analysis

We used the membrane protein extraction reagents from Thermo Fisher Scientific to isolate membrane proteins following the manufacturer's guidelines and established protocols [33]. 20 µg of protein was loaded into each well and separated by 10% SDS-PAGE gel electrophoresis. Subsequently, proteins were transferred overnight onto Immobileon® membranes (Millipore). The membrane was then blocked in a solution of 5% low-fat milk powder in Tris-buffered saline with 0.05% Tween 20 (50 mM Tris-HCl and 150 mM NaCl; pH 7.5) for 1 h. Following the blocking step, the membrane was exposed to antibodies targeting Toll-like receptor (TLR) 2, TLR4, PI3K, p-AKT, NF-κB, or total AKT (Santa Cruz Biotechnology). Next, the membrane was incubated with horseradish peroxidase conjugated secondary antibodies and chemiluminescence was detected following the manufacturer's instructions. Protein concentrations were determined using ImageJ Gel Analysis Tool (National Institutes of Health) and expressed as relative density units normalized to the corresponding β-actin control.

2.8 Measurement of tissue injury markers

Albumin depleted plasma was achieved using a commercially available kit (Albumin Removal Kit; Thermo Fisher Scientific). Plasma concentrations of blood creatinine and alanine transaminase (ALT) were detected using commercially available products (Pointe Scientific).

2.9 Statistical analysis

For each assay, more than three independent experiments were performed. Statistical relevance was estimated using one-way analysis of variance (ANOVA) and Tukey's post hoc test. Statistical significance was accepted at p values < 0.05. Survival analysis of CLP-induced sepsis outcomes was carried out using Kaplan–Meier analysis.

3 Results

3.1 Protective effects of lupeol on CLP-induced sepsis

The severity of sepsis induced by CLP surgery is influenced by the specific location of cecal puncture, leading to mid- or high-grade mortality outcomes [26]. Figure 2A illustrates the survival rates in mice subjected to mid- or high-grade CLP. The data reveal that in cases of high-grade sepsis, all mice succumbed, while mid-grade sepsis exhibited a 40% survival rate, consistent with prior research findings [26]. To assess the impact of lupeol on CLP-induced sepsis, mid- and high-grade CLP surgeries were performed. Lupeol was administered at varying concentrations (0.5, 1, or 2 mg/kg), with 0.5% DMSO serving as a negative control and MA (2 mg/kg) as a positive control [23–25]. In mid-grade sepsis, administration of lupeol at 1 or 2 mg/kg resulted in 75% or 90% survival over 10 days, respectively (Fig. 2B). However, 0.5 mg/kg of lupeol did not alter survival compared to the control group (Fig. 2B). The positive control group (MA) exhibited a 95% survival rate (Fig. 2B). These findings suggest that lupeol at a concentration of 2 mg/kg protects mice from CLP-induced sepsis, hence this concentration was used in subsequent experiments. Subsequently, lupeol (2 mg/kg) was assessed in a high-grade sepsis model (resulting in 100% mortality, Fig. 2A). The survival rate significantly improved ($p < 0.01$) compared to the control group (Fig. 2C), indicating that lupeol (2 mg/kg) can safeguard mice from CLP-induced sepsis, even in a high-grade sepsis model. The mid-grade CLP model resulted in weight loss (reaching up to – 17.5%) compared to pre-CLP surgery, with recovery of up to 30.5% observed after treatment with 2 mg/kg lupeol compared to CLP surgery at 5 days after CLP.

3.2 Suppressive effects of lupeol on levels of inflammatory cytokines and renal inflammation in mice subjected to CLP surgery

We investigated the effects of lupeol on the production of inflammatory cytokines in mice subjected to CLP. Lupeol

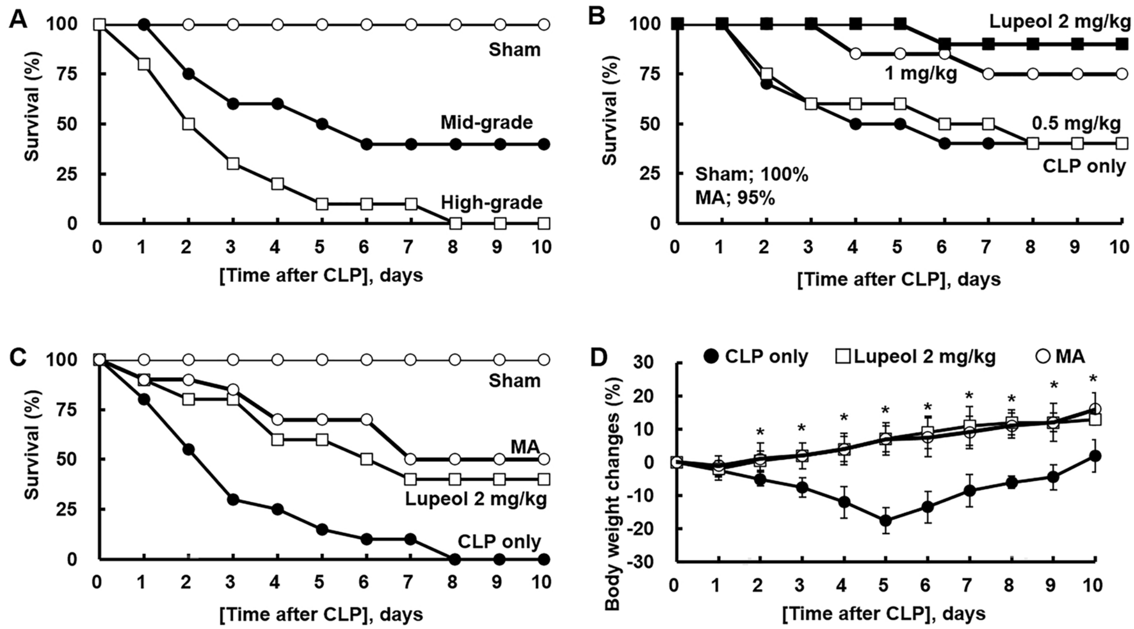

Fig. 2 Effects lupeol on the protection of mice from septic death by CLP. **A** Survival rates after induction of sepsis with various (mid or high) severity grades of CLP. **B**, **C** Each group comprised 20 mice and received intravenous injections of vehicle in 0.5% DMSO, MA (2 mg/kg), or lupeol (0.5, 1, or 2 mg/kg) 24 h post intermediate (**B**) or high-grade (**C**) CLP. **B** Black square, 2 mg/kg. White circle, 1 mg/kg. White square, 0.5 mg/kg; Black circle, CLP. Survival of challenged mice was monitored for 10 days. **D** Weight changes during the period in Figure (**B**) were measured. All values are presented as the mean ± SD of three separate experiments. CLP: cecal ligation and puncture, DMSO: dimethyl sulfoxide, MA: maslinic acid. $^*p < 0.01$ compared to the CLP group for lupeol or MA in Figure (**D**)

administration led to a reduction in cytokine levels, specifically TNF-α, IL-1β, and NO in the kidneys which is identified as the most sensitive organ to sepsis [34, 35] following high-grade CLP (Fig. 3A, B). Additionally, lupeol demonstrated anti-inflammatory effects by diminishing TNF-α and IL-1β levels in the liver (Fig. 3C). Acute inflammation occurs globally during the progression of sepsis and often leads to damage in vital organs, including MOF and its associated impairments [36]. We also investigated the therapeutic activity of lupeol against CLP-mediated organ damage. The concentration of ALT in serum, an indicator of liver damage, and the concentration of creatinine, an indicator of kidney damage, were significantly increased by the CLP surgery. Lupeol downregulated the increase of these markers (Fig. 3D). These findings support the anti-inflammatory properties of lupeol in vivo. Subsequently, we examined the effects of lupeol on renal inflammation using H&E staining of the kidneys. The results revealed that lupeol treatment significantly reduced inflammatory cell infiltration compared to the untreated group (Fig. 3E). This suggests that lupeol alleviates CLP-induced renal inflammation by suppressing the production of inflammatory cytokines.

3.3 Inhibitory effects of lupeol on TLR and NF-κB expression

Primary HUVECs were activated with HMGB1, a late sepsis mediator [17, 37], and subsequently treated with lupeol for 6 h. Measurement of TLR expression was conducted using Western blot analysis. Treatment with lupeol at concentrations of 13 or 25 μg/mL significantly decreased the expression of both TLR2 and TLR4 compared to the untreated control group (Fig. 4A). Moreover, lupeol treatment (13 or 25 μg/mL) suppressed the expression of NF-κB, the canonical downstream gene of the TLR pathway (Fig. 4A). Considering the average circulating blood volume for mice as 72 mL/kg [38], and the average weight of mice used in this study being 27 g with an average blood volume of 2 mL, the injected amount of lupeol (13 or 25 μg/mL) resulted in a maximum concentration of 1 or 2 mg/kg in the peripheral blood. Additionally, the production of TNF-α was significantly diminished when HUVECs were treated with lupeol (Fig. 4B), indicating that lupeol effectively suppressed TLR activation following HMGB1 stimulation. In summary, lupeol exhibited inhibitory effects on cytokine production and NF-κB expression in this experimental setup.

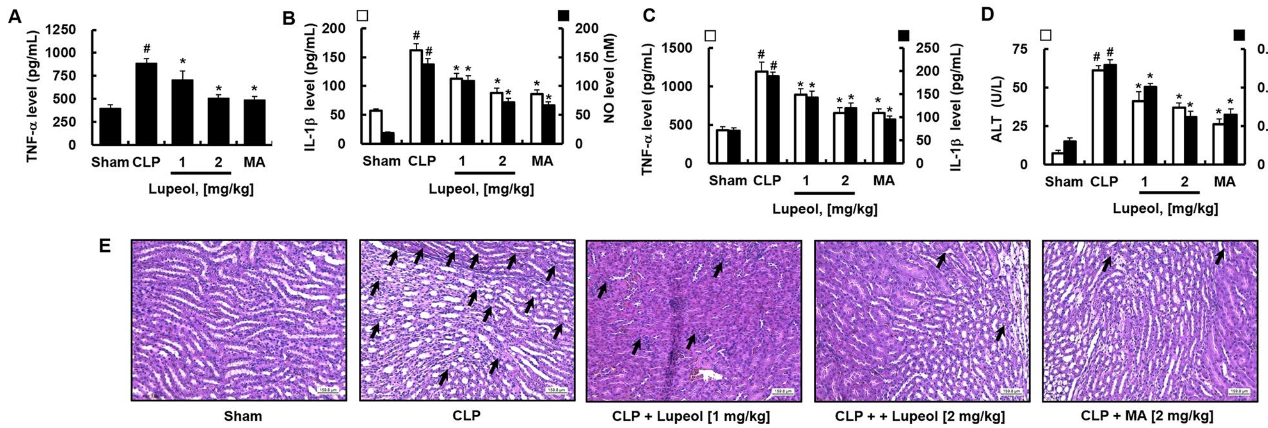

Fig. 3 Effects of lupeol on the levels of inflammatory cytokines and renal or hepatic inflammation in CLP. Each group consisted of five mice and received intravenous injections of vehicle in 0.5% DMSO, MA (2 mg/kg), or lupeol (1 or 2 mg/kg) 24 h post high-grade CLP. After 72 h of CLP, kidneys (**A**, **B**) or liver (**C**) were homogenized and used for the measurement of TNF-α (**A**) or IL-1β (**B**) in the kidneys or liver. **D** Inhibitory effects of lupeol on plasma levels of ALT or creatinine (n = 5). All values are presented as the mean ± SD of three separate experiments. $^{\#}p < 0.01$ vesus control group, or $^{*}p < 0.01$ versus CLP group. **E** Kidney tissues were stained with H&E. Arrows indicate infiltration of leukocytes. Microscopic images of kidney tissues at 72 h after CLP are shown (original magnification, 200×). Images represent three independent experiments. Scale bar: 160 µm. CLP: cecal ligation and puncture, DMSO: dimethyl sulfoxide, MA: maslinic acid, TNF-α: tumor necrosis factor-α, IL: interleukin, NO: nitric oxide, ALT: alanine transaminase

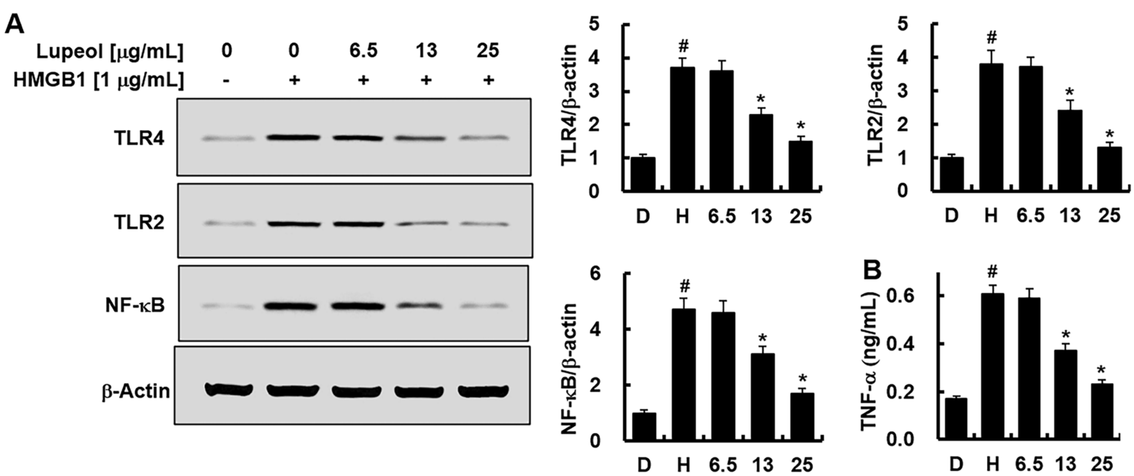

Fig. 4 Effects of lupeol on the expression of HMGB1-induced TLR/NF-κB. **A** HUVECs were treated with HMGB1 (1 µg/mL) for 6 h, followed by further activation with various concentrations of lupeol for an additional 6 h. Cell lysates were subjected to Western blot analysis to determine the protein levels of TLR2, TLR4, and NF-κB. Representative Western blot images (left) and quantification of relative intensity (right) for each gene. **B** Supernatants from (**A**) were used for TNF-α ELISA. All values are presented as the mean ± SD of three separate experiments. HMGB1: high mobility group box1, TLR: Toll-like receptor, NF-κB: nuclear factor-κB, HUVECs: human umbilical vein endothelial cells, TNF-α: tumor necrosis factor-α, DMSO: dimethyl sulfoxide, D: 0.5% DMSO is the vehicle control, H: HMGB1. $^{\#}p < 0.01$ versus DMSO or $^{*}p < 0.01$ versus HMGB1 group

3.4 Effects of lupeol on cell survival in vitro

Previous research has demonstrated the anti-tumor properties of lupeol, highlighting its effectiveness in modulating anti-inflammatory responses and inducing cell death by altering the expression levels of BCL-2, BAX, and the PI3K-AKT-mTOR signaling pathway in cancer cells [6]. Hence, we sought to investigate whether lupeol could enhance the survival of HUVECs during conditions simulating HMGB1-induced sepsis. To assess this, we conducted cell viability assays after activating HMGB1 with and without lupeol treatment. HMGB1 activation reduced HUVEC viability, while treatment with lupeol at concentrations of 13 or 25 µg/mL increased cell viability (Fig. 5A). This suggests that lupeol treatment enhances cell viability under HMGB1-induced septic conditions. However, lupeol alone did not

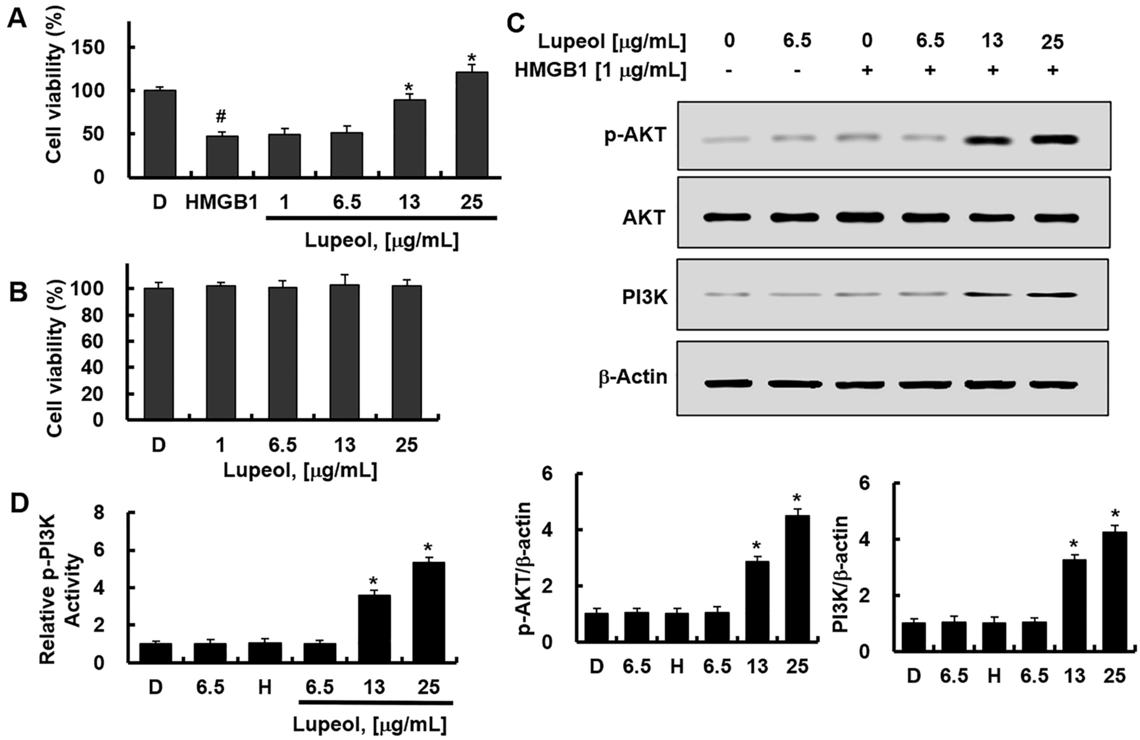

Fig. 5 Effects of lupeol on the cell survival through induction of PI3K/AKT signaling in HMGB1-mediated severe inflammatory response. HUVECs were activated with HMGB1 (1 μg/mL) with (**A**) or without (**B**) for 6 h, followed by further treatment with various concentrations of lupeol for an additional 6 h. (**A**, **B**) Cell viability was assessed by MTT assay. Cell lysates were subjected to Western blot analysis to estimate the protein levels of PI3K, p-AKT, and total AKT (**C**), or subjected to ELISA for phosphorylated PI3K (**D**). Representative Western blot images (left) and quantification of relative intensity (right) for each gene (**C**). All values are presented as the mean ± SD of three separate experiments. PI3K: phosphoinositide 3-kinase, HMGB1: high mobility group box1, HUVECs: human umbilical vein endothelial cells, DMSO: dimethyl sulfoxide, D: 0.5% DMSO is the vehicle control, H: HMGB1. $^{\#}p < 0.01$ versus DMSO or $^{*}p < 0.01$ versus HMGB1 group

affect the cell viability of HUVECs (Fig. 5B). Regarding PI3K/AKT signaling, PI3K levels remained unchanged following HMGB1 activation but were significantly elevated after lupeol treatment (Fig. 5C). Additionally, increased expression of p-AKT (Fig. 5C) and p-PI3K (Fig. 5D) was observed after lupeol treatment compared to the non-activated control group. However, total AKT levels remained unchanged (Fig. 5C), indicating that the enhanced cell viability induced by lupeol is associated with the activation of the PI3K/AKT pathway.

4 Discussion

Sepsis is a highly diverse syndrome characterized by an uncontrolled immune response to infection, often resulting in organ failure [17, 39]. It can be triggered by various infections, including bacteria, viruses, and fungi, and stands as the primary cause of mortality in intensive care unit patients [11, 12]. Major inflammatory factors contributing to sepsis include TNF-α, IL-1β, and IL-6 [17, 39]. While Xigris was the sole FDA-approved drug for sepsis treatment, it was globally withdrawn from the market in 2011 by Eli Lilly due to unfavorable results in a septic shock test involving 1700 patients [40]. The withdrawal of Xigris has left a void in approved drugs for sepsis treatment, emphasizing the need for a novel and safe therapeutic approach. In this study, lupeol demonstrated inhibitory effects on NO production and downregulated levels of inflammatory cytokines, including TNF-α and IL-1β, in mice with sepsis induced by CLP.

The administration of lupeol resulted in reduced levels of inflammatory cytokines such as TNF-α and IL-1β, lowered NO levels, and alleviated renal inflammation post-CLP surgery. These findings suggest that lupeol possesses potent anti-inflammatory properties. Moreover, in human endothelial cells stimulated with HMGB1 protein, lupeol decreased TLR4 and TNF-α expression. Additionally, lupeol activated the PI3K/AKT signaling pathway, which led to enhanced cell viability. These molecular mechanisms indicate that lupeol not only mitigates inflammatory responses but also strengthens cellular resilience against septic challenges. The study's outcomes contribute significantly to sepsis treatment by highlighting lupeol's potential therapeutic advantages. Lupeol's ability to enhance cell survival, reduce inflammatory cytokine production, and protect against CLP-induced sepsis-associated lethality and mortality suggests its role as a promising candidate for severe inflammatory conditions linked to sepsis. This research provides valuable insights into the mechanisms underlying lupeol's anti-sepsis effects, paving the way for further investigations and potential clinical applications of lupeol in treating sepsis and related inflammatory diseases.

When exploring the potential mechanism by which lupeol regulates the production of inflammatory cytokines, we evaluated the expression of TLR-2, TLR-4, and NF-κB in HUVECs. TLRs play a crucial role in modulating the production of inflammatory cytokines during the septic response, with early-stage expression of TLR-2 and TLR-4 correlating with mortality [41, 42]. NF-κB functions as a critical factor in TLR signal transduction and influences the pathogenesis of pneumococcal sepsis [43]. Our findings revealed that the septic state induced by HMGB1 enhances the expression of TLR-2, TLR-4, and NF-κB, while lupeol treatment attenuates this response. Lupeol therapy not only alleviated renal inflammation in CLP-induced sepsis but also resulted in higher survival rates compared to the untreated control group. Therefore, lupeol demonstrated beneficial effects in reducing TLR-mediated inflammation and subsequent sepsis induced by CLP surgery. These results underscore the protective effect of lupeol against CLP-induced sepsis.

The PI3K/AKT signaling pathway is highly crucial in promoting cell survival [44]. In sepsis, activation of HMGB1 leads to cell death, but treatment with lupeol enhances cell survival by increasing the expression of PI3K/p-AKT. Consequently, the upregulation of the PI3K/AKT signaling pathway might strengthen the resilience of cells against severe inflammatory reactions induced by HMGB1. Furthermore, blocking the PI3K/AKT pathway increases the production of cytokines and chemokines, suggesting that activation of the PI3K/AKT pathway could be advantageous in sepsis treatment. Therefore, the activation of the PI3K/AKT signaling pathway by lupeol in sepsis might not only inhibit inflammation but also promote cell survival. In this regard, lupeol could potentially play a dual role in suppressing the production of inflammatory cytokines and enhancing cell survival.

This study primarily focused on the effects of lupeol on inflammatory cytokines, cell survival, and the PI3K/AKT signaling pathway. However, sepsis involves a complex interplay of various molecular pathways, immune responses, and organ dysfunctions. Future studies might explore other mechanisms through which lupeol exerts its effects in sepsis. In addition, while this study provides valuable insights into the potential therapeutic benefits of lupeol in sepsis, there are several limitations to consider. These include the specific model used (CLP-induced sepsis in mice), the challenges of translating findings to human sepsis, the need for further dose optimization and safety studies, consideration of other molecular pathways involved in sepsis, and the inherent differences between mouse models and human patients. Additional research, including preclinical and clinical trials, would be necessary to validate the efficacy and safety of lupeol as a treatment for human sepsis.

The group treated with lupeol demonstrated an elevated survival rate and reduced mortality in contrast to the untreated control group. To summarize, lupeol enhances cell survival, diminishes the production of inflammatory cytokines, and thereby mitigates the lethality and mortality linked to sepsis. Lupeol holds the potential to offer therapeutic advantages for severe inflammatory diseases associated with sepsis by fortifying immunity.

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Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval This experimental protocol was approved by the Institutional Animal Care and Use Committee of Kyungpook National University before the start of the study (IRB No. KNU 2022-113). Jong-Sup Bae is an Editor of Biotechnology and Bioprocess Engineering. Editorial Board Editor status has no bearing on editorial consideration.

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